

The advances in electronics have largely made DSL a successful, faster and cheaper technology, though digging trenches in the ground for new copper or fibre-optic cables remains expensive. All types of DSL utilise very complex digital signal processing to overcome the inherent limitations of the existing copper pair. Until the late 1990s, digital signal processors for DSL were very expensive. But the rapid development in very large-scale integration (VLSI) technology has significantly lowered the cost of signal processors that supported DSL as a commercially successful technology.

DSL technology

Digital subscriber line (DSL) is referred to as a broadband technology because it is an 'always-on' data connection that is able to support interactive services including Internet access. It supports the minimum download speed of 256 kbps to an individual subscriber from the point of presence (POP) of the service provider where multiple such individual broadband connections are aggregated. The subscriber is able to access these interactive services including the Internet through this POP.

DSL technology is a modem technology that uses existing twisted-pair telephone lines to transport high-bandwidth data, such as multimedia and video, to service subscribers. xDSL services, a family of technologies, are dedicated, point-to-point, public network access over twisted-pair copper wire on the local loop between an Internet service provider's (ISP's) central office and the customer site, or on local loops created either intra-building or intra-campus.

In order to provide DSL connections, the existing telephone network is utilised and a network element called digital subscriber line access multiplexer (DSLAM) is installed at the central office/telephone exchange along with a modem placed in subscribers' home or workplace. The DSLAM works like a concentrator. It provides multiple DSL connections for access to the bandwidth available to the DSLAM

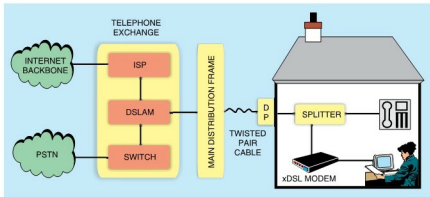


Fig. 1: A typical xDSL set-up

from the Internet backbone. The connection to the subscriber is then given from DSLAM via copper lines. The DSL modem performs the task of converting the DSL physical layer signal into a format that can be understood by a computer or any other equipment connected to it (Fig. 1).

The pair from the DSLAM is terminated into the main distribution frame (MDF), from where connections are provided to subscribers via twisted copper pair lines. These twisted pair lines terminate at DP box placed near a subscriber's home/workplace. The connection from DP box is first run to the splitter that sits in the customer's premises. The splitter, which is basically a low-pass filter, removes the plain old telephone system (POTS) signal (voice signal, 300-3400 Hz) from the incoming DSL signal. The high-frequency filtered DSL signal is then given to a DSL modem. Now the connection is taken out from Ethernet port of DSL modem and terminated to the subscriber's computer (Fig. 2).

The underlying technology of transport across DSL facilities is a high-frequency sinusoidal carrier modulation, which is an analogue-based signal transmission. Each end of a DSL circuit has a modem that modulates patterns of bits into certain high-frequency tones, representing that bit pattern, for transmission across the length of the facility. Tones received from the far-end modem are demodulated back into a corresponding bit pattern that the near-end modem retransmits, in true digital form as pulses of voltage,

to its interfaced equipment (such as a computer, router and switch).

Unlike traditional dial-up modems, which modulate bits into carrier that can fall only in the 300-3400Hz base-band (voice service), DSL modems modulate frequencies from 4 kHz to as high as 4 MHz. This frequency band separation enables DSL service and POTS to coexist on the same copper pair facility.

DSL technology divides the frequencies used in a single phone-line into two primary bands. The high-frequency band is used for data services and lower-frequency band (below 4 kHz) is utilised for voice. Ideally, DSL service provides 'always-on' connection to the customers to access the Internet. DSL service implemented with point-to-point over Ethernet (PPoE) does not support 'always-on' connection, but even in this case a DSL router/modem can automate the connection process.

Different xDSL technologies

The DSL technology is based on discrete multi-tone modulation technique and covers a number of similar yet competing forms of DSL (collectively termed as xDSL) including IDSL, HDSL, SHDSL, ADSL, RADSL, UDSL, Etherloop, VDSL and GDSL. xDSL is drawing significant attention from implementers and service providers because it promises to deliver high-bandwidth data rates to dispersed locations with relatively small changes to the existing telco infrastructure.

These technologies are differenti-